Ordinary Kummer sign recovery: next interface

Source-bound written derivation

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Derived component. Under the ordinary chart identity, the source delta-lift loop has at most two distinct candidates, related by simultaneous negation. Its present assertion allows four. This sharper bound is algebraic; no historical counter is changed. Let $u = X^2 + a_1X + a_0$, $v = b_0 + b_1X$ and $f \bmod u = R_0 + R_1X$. Validity gives

$$b_0^2 - a_0b_1^2 = R_0, \qquad 2b_0b_1 - a_1b_1^2 = R_1.$$

The native decoded chart intends to supply $S = b_1^2$. If $S \neq 0$, the two roots $\pm t$ force $b_0 = (R_1 + a_1S)/(2b_1)$, yielding opposite pairs. If $S = 0$, then $b_1 = 0$ and $b_0^2 = R_0$; again there are at most two opposite pairs. This works over any field of characteristic not2, with no $a_0 \neq 0$ or squarefree-u requirement for this local argument.

Orientation datum. The ordered pair $\pi(D), \pi(D+G)$ fixes the sign when the relevant quotient fibers are $\{D, -D\}$ and $2G \neq 0$. Both opposite signs passing the neighbour test would force either $2D = 0$ (identical candidates) or $2G = 0$. The chart identity supplying u, S, complete neighbour-chart fibers and primitive correctness remain concrete obligations.

Relation interface. For distinct split roots, signed recovery supplies $y_i = v(x_i)$ and the graph equation

$$s_0L(x_0) + s_1L(x_1) = aG + bQ.$$

Replacing D by $-D$ requires negating labels as well as node signs. The adjacent pair or two candidate signs supply one relation, not two independent rows. Native deduplication uses $\min(D, -D)$; discarded duplicates still incur recovery work.

Repair distinction. $M^2 = cI$ proves nonzero preservation with bounded row-fold arithmetic. The additional geometric equation $L_T\pi(P) = \pi(P + T)$ and $2T = 0$ explains why applying the repair twice returns the intended sum. Even identity matrices satisfy the square-scalar condition, so it cannot identify a designated nonzero torsion translation by itself.

Evidence. Three native source files match published5cd0a45205. Eleven fixed native algebra assertions passed, including wrong independent sign, two-torsion-anchor ambiguity and wrong graph orientation. No curves were computed. This is a written derivation awaiting independent review, not a new Lean theorem or measurement. Detailed equations, source lines and missing obligations accompany this PDF.